

TIGHT MONEY AND INFLATION

Further Results

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The Sargent–Wallace results on inflationary effects of monetary restraint are considered in a Sidrauski money growth model when the utility function is unrestricted except for separability. Temporary monetary tightening will eventually lead to higher inflation when the deficit is fixed only if the elasticity of money demand with respect to the money growth rate is less than unity. It is also shown that for inflation to rise immediately further requires the elasticity with respect to the nominal interest rate to be no less than unity.

1. Introduction

Sargent and Wallace (1981) have recently argued that when the government deficit (exclusive of debt service) is held constant, temporary monetary restraint may eventually lead to higher inflation. They even show cases (termed ‘spectacular’) where temporary tightening of monetary policy causes an immediate and permanent increase in inflation. The results arise because smaller reliance on money creation implies greater bond issuance and growing debt service. An upper limit on public willingness to hold bonds means that the tight monetary policy must eventually be abandoned. When it is, money creation must finance higher debt service than would otherwise have been the case, implying higher inflation.

Because the SW model has a number of non-standard features, the question has arisen as to the generality of their results. Liviatan (1984) considered this question by casting the analysis in a Sidrauski (1967) monetary growth model and showed that the spectacular result can be obtained in such a set-up. Liviatan’s excellent analysis was, however, limited by its assumption of a Cobb–Douglas utility function.

The purpose of this paper is to consider the SW policy experiment in a standard monetary growth model for more general utility functions (retaining separability across time and commodities). It turns out that the SW results will

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hold only in certain cases. Consideration of when these results do and do not hold will help to understand why their seemingly paradoxical results can occur.

The analysis presented here has more general applicability. As will be demonstrated, the steady state is not a saddle point: in the two-variable system, the linear approximation about the steady state has two unstable roots. (Technically, the steady state is a source.) Nonetheless, the usual dynamic analysis between steady states is possible, because of the nature of the policy experiment considered. The results therefore suggest that even if one is interested in analysis between steady states, one is not necessarily restricted to models with stable equilibria.

2. The basic model

For simplicity, we use notation and basic set-up similar to Liviatan's. The representative family maximizes the discounted sum of utility over an infinite horizon, where instantaneous utility is a function of per-capita consumption and real balances. Welfare derived from a path is

$$\int_0^{\infty} U(c(t), m(t))e^{-\delta t} dt. \quad (1)$$

There is a constant real per-capita flow endowment of $y(t)$, which is taxed at an exogenously fixed rate τ . Since transfers play no substantive role, they will be assumed to be zero. Bonds (per capita) b are indexed (as in SW), pay real gross interest at rate r , and are also taxed at rate τ . Income is divided between consumption and gross asset accumulation, the consumer's budget constraint being

$$\begin{aligned} c(t) + (\dot{b}(t) + nb(t)) + (\dot{m}(t) + (\Pi(t) + n)m(t)) \\ = (1 - \tau)(rb(t) + y(t)), \end{aligned} \quad (2)$$

where n is the rate of population growth, Π the rate of inflation, $\dot{x} = dx/dt$, and the t indices indicate which variables may vary over time. Total per-capita assets at time zero, a_0 , are given, yielding

$$m(0) + b(0) = a_0. \quad (3)$$

Optimal time paths for c , m , and b may be derived by maximizing (1) subject to (2) and (3). Defining $q(t)$ as the Lagrange multiplier on constraint (2), we obtain the first-order conditions (subscripts on U denoting partial

derivatives)

$$U_c - q = 0, \quad (4)$$

$$U_m - q(\Pi + n + \delta) = -\dot{q}, \quad (5)$$

$$q((1 - \tau)r - n - \delta) = \dot{q}, \quad (6)$$

plus, of course, eqs. (2) and (3). The time index has been omitted.

We add to this the aggregate national income identity

$$y = c + g, \quad (7)$$

where g denotes real government expenditure per capita, assumed fixed as in SW.

To analyze policy experiments, we must derive time paths for b and m . To do this, we restrict the utility function to be separable in c and m . By (4), q is then a function of c alone, which by (7) is fixed. Then q is constant over time, so that (6) yields $(1 - \tau)r = n + \delta$, r therefore being exogeneously fixed. Using these results, we are left with eqs. (2) and (5), which become

$$\dot{m} + \dot{b} = \delta b + (1 - \tau)y - (\Pi + n)m - c, \quad (8)$$

$$U_m = q(\Pi + n + \delta), \quad (9)$$

where the term in parentheses can be written $\Pi + (1 - \tau)r$, or the net nominal interest rate i . Note that U_m is a function of m alone. To these, we add an equation defining the rate of change of m ,

$$\dot{m} = (\mu - \Pi - n)m, \quad (10)$$

where μ is the rate of growth of the nominal money supply. Substituting (10) into (8) we obtain

$$\dot{b} = \delta b - \mu m + [(1 - \tau)y - c], \quad (11)$$

where the term in square brackets, which is simply the deficit, is fixed. Call it D . Substituting (9) into (10) we obtain

$$\dot{m} = (\mu + \delta - U_m/q)m. \quad (12)$$

Eqs. (11) and (12) are the basic dynamic equations of the system, expressing the change in m and b as functions of m and b alone, for a given value of the policy parameter μ .

3. The Sargent–Wallace policy experiment

The SW experiment is as follows. Starting in steady state at t_0 , we reduce the money growth rate from μ_0 to μ' for a finite period of time, until t_1 . At t_1 policy is such that b and m are frozen at their (real) t_1 levels and μ is changed to whatever level is necessary to make $m(t_1)$ and $b(t_1)$ a steady state. The monetary authority undertakes no open-market operations, so that discrete changes in m and b are ruled out. The finite duration of μ' is assumed known to the public as of t_0 . (The value of μ and the permanent jump to a new μ at t_1 are given characteristics of the problem, not themselves arising from welfare maximization.)

To analyze the effects of a temporary reduction in μ , we consider phase diagrams. For given μ , the $\dot{b} = 0$ locus is defined by

$$b = (\mu/\delta)m - D/\delta. \quad (13)$$

This locus positively sloped in $m - b$ space, b rising to the right of the locus, falling to the left. The $\dot{m} = 0$ locus is defined by

$$U_m[m]/q = \mu + \delta, \quad (14)$$

where m is falling below this horizontal locus, rising above it. The motion of the system may then be represented by the phase diagram in fig. 1. The steady state is a point of unstable equilibrium. Decreases in μ shift both loci up. Whether the new steady state point (at lower μ) is one with higher or lower b depends on the relative size of the shifts of the two loci. In fig. 2 we draw the locus of steady state combinations of m and b (each for a different μ), denoted SS , when the shift in the $\dot{b} = 0$ locus is large relative to the shift in the $\dot{m} = 0$ locus.

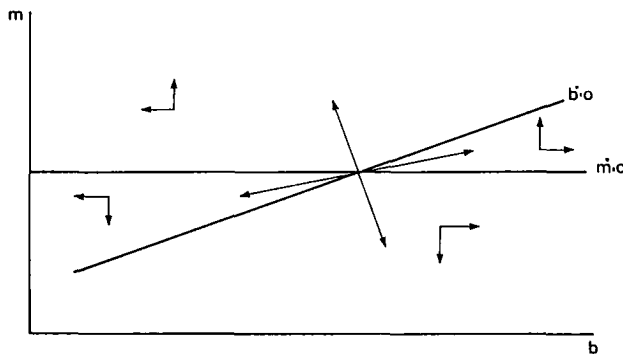


Fig. 1. Phase diagram for motion of the system.

The full time path may now be analyzed. Before t_0 , the economy is at steady state E_0 . At t_0 , μ_0 is lowered to μ' , so the motion is that corresponding to the loci for μ' , as in fig. 2. Since it is known at t_0 that, at t_1 , μ_1 will be chosen so that $m(t_1)$ and $b(t_1)$ are steady state values, we jump at t_0 to a point which implies that we will just arrive to a point on SS at t_1 . Then μ_1 will be that rate of monetary growth such that the $\dot{b} = 0$ and $\dot{m} = 0$ loci intersect at that point. At t_0 , b is fixed, but m can jump due to a price level jump. For example, we jump to a point such as E' in fig. 2, arriving at E_1 at t_1 , μ_1 being set to make this a steady state. (Ignore, for now, the path E_0T .) For E_1 below E_0 , m_1 is less than m_0 , so that μ_1 is greater than μ_0 .

Note that if open-market operations were undertaken, both m and b could jump at t_0 , so that we could jump immediately to a new steady state. Any change in μ could be permanent, Π moving in the same direction. The absence of open-market operations is therefore crucial for the SW result to hold. It alone will not, however, be sufficient.

The characteristics of both the new steady state and the transition will depend on the nature of the SS line. Its equation, obtained by solving μ out of (13) and (14), is

$$b = mU_m/q\delta - m - D/\delta. \quad (15)$$

In the reference case where SS is vertical, the fall in μ induces a jump up in m , which brings us immediately to a new steady state. (E' is on SS .) The stock of bonds remains constant and $\mu_1 = \mu' < \mu_0$. We remain at E' and the SW result doesn't hold.

Mathematically, a vertical SS means that (13) and (14) are satisfied for only one value of b , so [from (13)] μm must be constant across steady states. $U(c, m)$ is such that the elasticity of money demand with respect to μ is one.

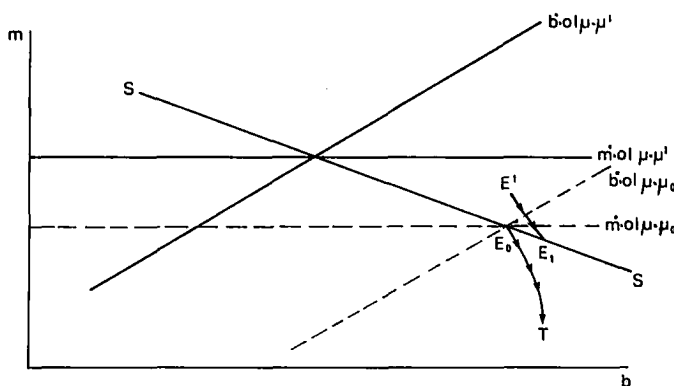


Fig. 2. Effect of a temporary fall in the growth rate of money.

Intuitively, since revenue from money creation is independent of μ , a decrease in μ puts no additional burden on bond finance, and the tightening of monetary policy can be permanent.

A steady state demand elasticity with respect to μ greater than one means SS will be positively sloped. [From (9), U_{mm} will be smaller than in the reference case, so that db/dm in (15) will be positive.] A fall in μ at t_0 requires m to jump up if we are to hit the SS locus at t_1 . At E_1 , the new steady state, both b and m are higher than their initial positions, so, by (9), the inflation rate after t_1 is lower than before t_0 . The SW result will not hold, even in weak form. Intuitively, though μm must be higher in the new steady state (since b is higher), the greater than unitary elasticity means that μ and Π can be lower.

What about cases where the elasticity of money demand with respect to μ is less than one (SS negatively sloped), so that higher steady state seignorage revenue requires higher μ ? Will temporary monetary restraint necessarily imply an immediate increase in inflation? From (9) this is equivalent to asking whether a temporary fall in μ implies that $m(t)$ is less than m_0 for all $t \geq t_0$. Examination of the phase diagram in fig. 2 [or solution of the differential equation system (11) and (12) imposing the boundary condition that we are on SS at t_1] indicates that $\dot{m} < 0$ between t_0 and t_1 . We must therefore ask how m jumps at t_0 . If it jumps up (as in fig. 2), Π will initially fall in response to reduced μ , rising above its pre- t_0 value only when m falls below m_0 (as it eventually must do when SS slopes down). If m does not jump or jumps down at t_0 , the inflation rate will be uniformly above its initial value, and the 'spectacular' SW result will hold.

This may be easily analyzed. Combining (8) and (9) the change in total assets may be written

$$\dot{a} = \delta(b + m) - mU_m/q + D. \quad (16)$$

(Note that the second right-hand-side term is simply $-im$.) Comparing (16) and (15), one sees that the locus of points such that $\dot{a}$ equals zero is identical to the SS locus. Therefore below SS , $\dot{a}$ is falling; above, $\dot{a}$ is rising; while combinations on SS imply $\dot{a} = 0$ along the optimal path. Equivalently, the slope of the path from E' to E_1 will be less than, equal to, or greater than one in absolute value depending on whether we are above, on, or below SS .

This characteristic of the solution indicates that the jump in m depends on the slope of SS . Suppose it has slope less than one in absolute value (SS flat). Were there no jump in m at t_0 , the path would start on SS and have slope equal to -1 . It would therefore be below SS immediately afterwards. Since the slope of the path is always above one in absolute value in this region, the path will stay below SS and never hit it. (Consider E_0T in fig. 2.) The same would be true if m jumps down. Only if m jumps up, so the slope of the path is less

than one but greater than that of SS in absolute value, will the path cross SS at some time after t_0 . That is, at some $t > t_0$, $\dot{a} = 0$, which is necessary for steady state. The necessity of crossing precisely at t_1 determines the size of the jump.

Similarly, if SS has slope greater than one, a jump up in m at t_0 will mean the path never crosses SS . A jump down is necessary to satisfy the boundary condition. Only if SS has slope equal to -1 will there be no jump. A path along which $\dot{a} = 0$ stays on SS and can satisfy the boundary condition at t_1 . This will be true for a Cobb–Douglas utility function, the case analyzed by Liviatan.

To demonstrate this more formally let us write im as I and $\dot{a}$ as A in (16), yielding

$$A = \delta a - I + D. \quad (16')$$

Differentiating (16') with respect to time, we obtain

$$\dot{A} = \delta A - (dI/dm)(dm/dt). \quad (17)$$

[The functional relation of i and m in (9) allows $\dot{I}$ to be written as in (17).] What must happen to A at t_0 to ensure that $A = \dot{a} = 0$ at t_1 depends on the sign of dI/dm . If it is zero (as with a Cobb–Douglas utility function), $A = 0$ along the whole path. (In the SW example im was zero along the path, analogous to the no jump case.) If $dI/dm > 0$ throughout, so that $-(dI/dm)\dot{m}$ is positive from t_0 to t_1 , the boundary condition for A at t_1 clearly requires A to be negative at t_0 (but $\dot{A}$ to be positive). $A(t_1) < 0$ requires a jump below SS at t_0 . Conversely, if SS has slope greater than one in absolute value, so that $dI/dm < 0$ throughout, $-(dI/dm)\dot{m} < 0$ throughout the transition. $A(t_1) = 0$ requires $A(t_0) > 0$, so that m must jump above SS at t_0 .

The implications for the time path of Π in response to temporary monetary restraint are immediate. If the interest elasticity of money demand is unity (the Liviatan case), there will be no jump. Since m is negative, Π will immediately start rising, so that the 'spectacular' result will hold. It will also hold if the interest elasticity is greater than one, since m will jump down. Π will jump up immediately and stay permanently above its pre-tightening level. However, if the interest elasticity of money demand is less than unity, Π jumps down, rising above its pre-tightening level only after a finite period of time. The weak SW result will hold, but the spectacular one will not.

4. Summary

Temporary monetary tightening will *eventually* lead to higher inflation when government spending and taxes are held fixed only if the elasticity of m with

respect to the money growth rate is less than unity. For it to do so immediately further requires the elasticity with respect to the nominal interest rate to be greater than or equal to unity. Intuitively, with g and τ fixed, the discounted value of expenditures and tax revenue are fixed. Therefore, the discounted value of revenue from money creation must also be fixed. If less seignorage revenue is collected today, more must be collected tomorrow (yielding the rise in debt), where revenue at each instant is μm . Hence μ will eventually rise if the elasticity with respect to μ is less than unity and fall if it is greater. When the elasticity equals one, seignorage is independent of μ , and the temporary contraction is permanent. For inflation to increase immediately, an initial fall in wealth must be required for eventual stabilization to be consistent with individual optimization.

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